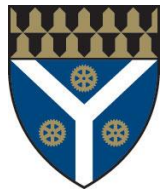




GPU-Accelerated Agent-Based Simulation for Cell Dynamics

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INTRODUCTION

Why artificial life + cancer?

Emergence: simple local rules → complex global patterns
(Conway's Life, Turing morphogenesis)

Evolutionary agents (NEAT, Polyworld) show decentralized adaptation

Tumors as evolving ecosystems: cells compete for nutrients, secrete signals, mutate

Goal: build a sandbox that starts with CA and energy sharing, then layers on chemistry, GRNs, and open-ended evolution

KEY FINDINGS

Local neural policies yield emergent cycles of growth and collapse; "energy wave" patterns and stable micro-colonies.

Chemistry + GRNs produce oxygen gradients, nutrient rings, and self-organizing cell clusters—tumor-like cores emerge naturally.



EXPERIMENT 1: ENERGY-CA WITH NEURAL POLICIES

Each grid-cell: alive/dead + continuous energy reservoir

Death → pulses energy into environment; birth → draws from 3×3 neighbor energy

EnergyNCA: tiny MLP per cell maps local energy + genome to a "demand"

Redistribution: normalize local demands, roll out energy packets, add 1% diffusion + uniform leak

Evolution: 3-way crossover + per-gene mutation (mutation-rate gene included)

EXPERIMENT 2: CHEMICAL GRID + GRNs

Chemicals: H₂O, CO₂, O₂, Sugar, ATP, Protein, Mitogen on one toroidal grid

Diffusion: per-species Laplacian (gases fast; sugar slow)

Reactions: nameless list (respiration, photosynthesis, biosynthesis, mitogen synth/degrad)

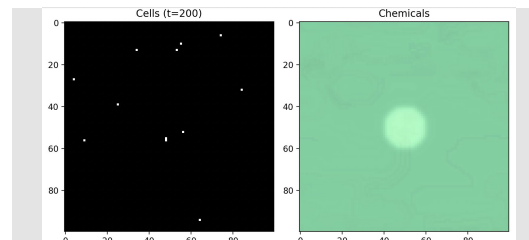
- background (global) + cell-facilitated at genome-controlled rates (ATP cost)

Cells = internal "channels":

- Movement (M_i), Transport (T_i), Reaction (R_i) channels per chemical/reaction
- Produced by GRN at ATP cost, decay over time

GRN: small sigmoid layer mapping local fields + channels → target channels

Mitosis: when Mitogen ≥ threshold, split all local chemicals 50/50 into a random empty neighbor + mutate GRN



Snapshot of the simulation at t=200. White dots are live cells; green field shows chemical concentrations, with the bright spot marking a mitogen pulse.

IMPLICATIONS & FUTURE WORK

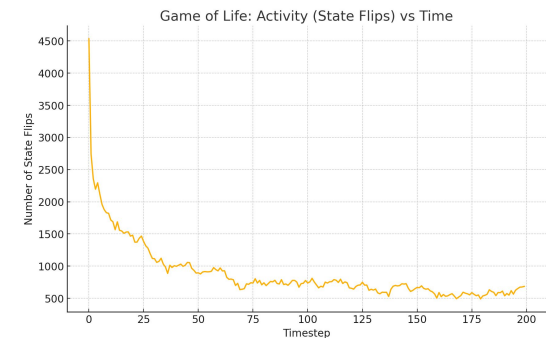
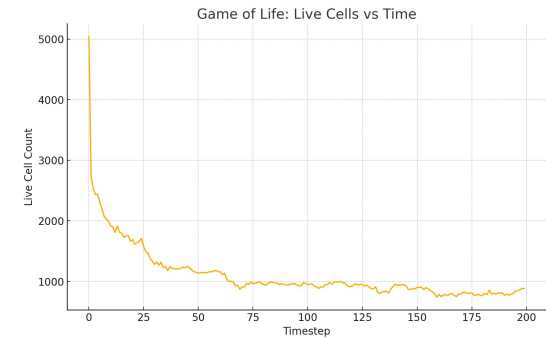
Cancer modeling: hypoxic cores, nutrient competition, signaling-driven invasion

Synthetic biology: in silico testbed for engineered GRNs and metabolic consortia

Open-ended evolution: no global fitness → continuously novel phenotypes

Extensions:

- Add immune-like agents or drug chemicals
- 3D grid for tissue volumetrics
- GPU acceleration (CuPy/JAX) for larger worlds



CONCLUSIONS

Emergent complexity arises from layering energy, neural policies, and chemistry

Decentralized rules + evolution → lifelike behavior (metabolism, movement, division)

Sandbox potential: bridge ALife theory and practical cell-dynamics modeling for cancer